Kth Smallest

Difficulty: **Medium** Accuracy: **35.17%** Submissions: **736K+** Points: **4**

Average Time: **25m**

Given an integer array **arr[]** and an integer **k**, your task is to find and return the **k**th **smallest** element in the given array.

Note: The kth smallest element is determined based on the sorted order of the array.

Examples :

Java (21) Start Timer

```java
import java.util.*;

class Solution {
    public static int kthSmallest(int[] arr, int k) {
        PriorityQueue<Integer> pq = new PriorityQueue<>();
        for (int x : arr) {
            pq.add(x);
        }
        for (int i = 1; i < k; i++) {
            pq.poll();
        }
        return pq.peek();
    }

    public static void main(String[] args) {
        int[] arr1 = {10, 5, 4, 3, 48, 6, 2, 33, 53, 10};
        int k1 = 4;
        System.out.println(kthSmallest(arr1, k1));

        int[] arr2 = {7, 10, 4, 3, 20, 15};
        int k2 = 3;
        System.out.println(kthSmallest(arr2, k2));
    }
}
```

Custom Input Compile & Run Submit

Output Window

Compilation Results Custom Input Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed	Attempts : Correct / Total
1121 / 1121	**2 / 2**

Minimize the Heights II

Difficulty: **Medium** Accuracy: **15.06%** Submissions: **770K+** Points: **4**

Average Time: **25m**

Given an array **arr[]** denoting heights of **n** towers and a positive integer **k**.

For **each** tower, you must perform **exactly one** of the following operations **exactly once**.

- **Increase** the height of the tower by **k**
- **Decrease** the height of the tower by **k**

Java (21) Start Timer

```java
import java.util.*;

class Solution {
    public static int getMinDiff(int[] arr, int k) {
        int n = arr.length;
        Arrays.sort(arr);

        int ans = arr[n - 1] - arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] - k < 0) continue;

            int minVal = Math.min(arr[0] + k, arr[i] - k);
            int maxVal = Math.max(arr[n - 1] - k, arr[i - 1] + k);

            ans = Math.min(ans, maxVal - minVal);
        }

        return ans;
    }

    public static void main(String[] args) {
        int[] arr1 = {1, 5, 8, 10};
        int k1 = 2;
        System.out.println(getMinDiff(arr1, k1));

        int[] arr2 = {3, 9, 12, 16, 20};
        int k2 = 3;
        System.out.println(getMinDiff(arr2, k2));
    }
}
```

Custom Input Compile & Run Submit

Output Window

Compilation Results Custom Input Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed	Attempts : Correct / Total
1115 / 1115	**1 / 2**

Minimum Jumps

Difficulty: **Medium** Accuracy: **11.91%** Submissions: **1.1M** Points: **4**

You are given an array **arr[]** of non-negative numbers. Each number tells you the **maximum number of steps** you can jump forward from that position.

For example:

- If **arr[i] = 3**, you can jump to index **i + 1**, **i + 2**, or **i + 3** from position **i**.
- If **arr[i] = 0**, you **cannot jump forward** from that position.

Output Window

Compilation Results Custom Input Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed	Attempts : Correct / Total
1120 / 1120	**2 / 2**

Java (21) Start Timer

```java
class Solution {
    public static int minJumps(int[] arr) {
        int n = arr.length;
        if (n <= 1) return 0;
        if (arr[0] == 0) return -1;

        int jumps = 1;
        int maxReach = arr[0];
        int steps = arr[0];

        for (int i = 1; i < n; i++) {
            if (i == n - 1) return jumps;

            maxReach = Math.max(maxReach, i + arr[i]);
            steps--;

            if (steps == 0) {
                jumps++;
                if (i >= maxReach) return -1;
                steps = maxReach - i;
            }
        }
        return -1;
    }

    public static void main(String[] args) {
        int[] a1 = {1, 3, 5, 8, 9, 2, 6, 7, 6, 8, 9};
        System.out.println(minJumps(a1));

        int[] a2 = {1, 4, 3, 2, 6, 7};
        System.out.println(minJumps(a2));
```

Custom Input Compile & Run Submit

leetcode.com/problems/merge-intervals/description/

Problem List | Submit | Premium

Description | Editorial | Solutions | Submissions

56. Merge Intervals

Solved

Medium | Topics | Companies

Given an array of `intervals` where `intervals[i] = [start_i, end_i]`, merge all overlapping intervals, and return *an array of the non-overlapping intervals that cover all the intervals in the input.*

Example 1:

```
Input: intervals = [[1,3],[2,6],[8,10],[15,18]]
Output: [[1,6],[8,10],[15,18]]
Explanation: Since intervals [1,3] and [2,6] overlap, merge them into [1,6].
```

Example 2:

```
Input: intervals = [[1,4],[4,5]]
Output: [[1,5]]
```

24.3K | 263 | 396 Online

Code

Java | Auto

```java
import java.util.*;

class Solution {
    public int[][] merge(int[][] intervals) {
        Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
        List<int[]> res = new ArrayList<>();

```

Saved | Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 1 ms

Case 1 | Case 2 | Case 3

Input

intervals =

[[1,3],[2,6],[8,10],[15,18]]

Merge Without Extra Space

Difficulty: **Medium** Accuracy: **32.01%** Submissions: **326K+** Points: **4**

Average Time: **20m**

Given two sorted arrays **a[]** and **b[]** of size **n** and **m** respectively, the task is to merge them in sorted order without using any **extra space**. Modify **a[]** so that it contains the first **n** elements and modify **b[]** so that it contains the last **m** elements.

Examples:

Input: a[] = [2, 4, 7, 10], b[] = [2, 3]

Output Window

Compilation Results | Custom Input | Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed	Attempts : Correct / Total
1111 / 1111	**1 / 1**

Java (21) | Start Timer

```java
            while (j < n + m) {
                if (i < n && j < n) {
                    if (a[i] > a[j]) {
                        int temp = a[i];
                        a[i] = a[j];
                        a[j] = temp;
                    }
                } else if (i < n && j >= n) {
                    if (a[i] > b[j - n]) {
                        int temp = a[i];
                        a[i] = b[j - n];
                        b[j - n] = temp;
                    }
                } else {
                    if (b[i - n] > b[j - n]) {
                        int temp = b[i - n];
                        b[i - n] = b[j - n];
                        b[j - n] = temp;
                    }
                }
                i++;
                j++;
            }

            if (gap == 1) break;
            gap = (gap + 1) / 2;
        }
    }
}
```

Custom Input | Compile & Run | Submit

leetcode.com/problems/merge-intervals/description/

Problem List | Submit | Premium

Description | Accepted | Editorial | Solutions | Submissions

56. Merge Intervals

Solved

Medium | Topics | Companies

Given an array of `intervals` where `intervals[i] = [start_i, end_i]`, merge all overlapping intervals, and return *an array of the non-overlapping intervals that cover all the intervals in the input.*

Example 1:

```
Input: intervals = [[1,3],[2,6],[8,10],[15,18]]
Output: [[1,6],[8,10],[15,18]]
Explanation: Since intervals [1,3] and [2,6] overlap,
merge them into [1,6].
```

Example 2:

```
Input: intervals = [[1,4],[4,5]]
Output: [[1,5]]
```

24.3K | 263 | 397 Online

Code

Java | Auto

```java
import java.util.*;

class Solution {
    public int[][] merge(int[][] intervals) {
        Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
        List<int[]> res = new ArrayList<>();

```

Saved | Ln 25, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

intervals =

[[1,3],[2,6],[8,10],[15,18]]

Common in 3 Sorted Arrays

Difficulty: **Easy** Accuracy: **22.16%** Submissions: **439K+** Points: **2**

Given three sorted arrays in **non-decreasing** order, print all common elements in **non-decreasing** order across these arrays. If there are no such elements return an empty array. In this case, the output will be -1.

Note: can you handle the duplicates without using any additional Data Structure?

Examples :

Java (21) Start Timer

```java
class Solution {
    public static ArrayList<Integer> commonElements(int[] arr1, int[] arr2,
        ArrayList<Integer> res = new ArrayList<>();
        int i = 0, j = 0, k = 0;

        while (i < arr1.length && j < arr2.length && k < arr3.length) {
            if (arr1[i] == arr2[j] && arr2[j] == arr3[k]) {
                if (res.size() == 0 || res.get(res.size() - 1) != arr1
                    res.add(arr1[i]);
                }
                i++;
                j++;
                k++;
            } else if (arr1[i] < arr2[j]) {
                i++;
            } else if (arr2[j] < arr3[k]) {
                j++;
            } else {
                k++;
            }
        }

        if (res.size() == 0) {
            res.add(-1);
        }

        return res;
    }
}
```

Custom Input Compile & Run Submit

Output Window

Compilation Results Custom Input Y.O.G.I. (AI Bot)

Compilation Error

Ask Yogi Bot

```
./GFG.java: 31: error: incompatible types: List<Integer> cannot be converted to int[]
        List<Integer> res = ob.commonElements(arr, brr, crr);
                                                        ^
```

Factorials of large numbers

Difficulty: **Medium** Accuracy: **36.57%** Submissions: **177K+** Points: **4**

Average Time: **20m**

Given an integer **n,** find its factorial. Return a list of integers denoting the digits that make up the factorial of n.

Examples:

Input: n = 5

Output: [1, 2, 0]

Java (21) Start Timer

```java
import java.util.*;

class Solution {
    static ArrayList<Integer> factorial(int n) {
        ArrayList<Integer> res = new ArrayList<>();
        res.add(1);

        for (int x = 2; x <= n; x++) {
            int carry = 0;
            for (int i = 0; i < res.size(); i++) {
                int val = res.get(i) * x + carry;
                res.set(i, val % 10);
                carry = val / 10;
            }
            while (carry > 0) {
                res.add(carry % 10);
                carry /= 10;
            }
        }

        Collections.reverse(res);
        return res;
    }
}
```

Output Window

Compilation Results Custom Input Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed	Attempts : Correct / Total
1111 / 1111	**1 / 1**

Custom Input Compile & Run Submit

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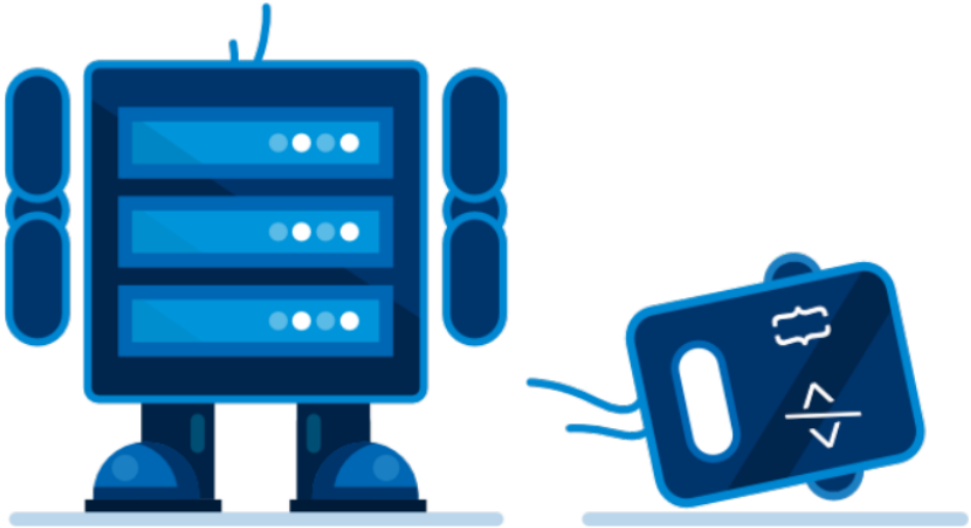

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Triplet Sum in Array

Difficulty: **Medium** Accuracy: **35.0%** Submissions: **360K+** Points: **4** Average Time: **15m**

Given an array **arr[]** and an integer **target**, determine if there exists a triplet in the array whose sum equals the given **target**.

Return **true** if such a triplet exists, otherwise, return **false**.

Examples:

Input: arr[] = [1, 4, 45, 6, 10, 8], target = 13

Java (21) | Start Timer

```java
import java.util.*;

class Solution {
    public boolean hasTripletSum(int[] arr, int target) {
        Arrays.sort(arr);
        int n = arr.length;

        for (int i = 0; i < n - 2; i++) {
            int l = i + 1, r = n - 1;
            int need = target - arr[i];

            while (l < r) {
                int sum = arr[l] + arr[r];
                if (sum == need) return true;
                if (sum < need) l++;
                else r--;
            }
        }
        return false;
    }
}
```

Custom Input | Compile & Run | Submit

Output Window

Compilation Results | Custom Input | Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed	Attempts : Correct / Total
1111 / 1111	**1 / 1**

Trapping Rain Water

Difficulty: **Hard** Accuracy: **33.14%** Submissions: **497K+** Points: **8** Average Time: **20m**

Given an array **arr[]** with non-negative integers representing the height of blocks. If the width of each block is 1, compute how much water can be trapped between the blocks during the rainy season.

Examples:

Input: arr[] = [3, 0, 1, 0, 4, 0 2]
Output: 10

Java (21) Start Timer

```java
class Solution {
    public int maxWater(int[] arr) {
        int n = arr.length;
        int left = 0, right = n - 1;
        int leftMax = 0, rightMax = 0;
        int water = 0;

        while (left < right) {
            if (arr[left] <= arr[right]) {
                if (arr[left] >= leftMax) {
                    leftMax = arr[left];
                } else {
                    water += leftMax - arr[left];
                }
                left++;
            } else {
                if (arr[right] >= rightMax) {
                    rightMax = arr[right];
                } else {
                    water += rightMax - arr[right];
                }
                right--;
            }
        }
        return water;
    }
}
```

Custom Input Compile & Run Submit

Output Window

Compilation Results Custom Input Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed	Attempts : Correct / Total
1111 / 1111	**1 / 1**